Week 11 — Multiple regression II: categories and interactions

Business Forecasting

My code (.R) Notes

Which airline lands latest?

Week 10 forecast the arrival delay at Mexico City airport from the departure delay and the scheduled time.
`flights$carrier` puts each flight in one of five groups:

- “Aeromexico”, “Volaris”, “Viva Aerobus”: the three Mexican airlines
- “US airline”: United, American and Delta
- “Other foreign”: every other foreign airline (Avianca, Copa, Air Canada, Iberia…)

An airline’s operations team wants to know where it stands: which airline lands latest on average, and which earliest?

1 · A regressor with several categories

One variable, five groups: how a regression reads it.

Exercise 1 — Mean delay by airline

Which airline has the highest mean arrival delay? Which has the lowest? How many minutes apart are they?

1. Compute the mean of `flights$arr_delay` for each of the five values of `flights$carrier`.
2. Compare each mean with the Aeromexico mean.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exercise 2 — The airline in a regression

`lm()` accepts a text column as a regressor. What do its coefficients mean?

1. Regress `flights$arr_delay` on `flights$carrier`.
2. Compare the intercept with the Aeromexico mean of Exercise 1.
3. Compare the coefficient on `carrierVolaris` with the gap between the Volaris mean and the Aeromexico mean.
4. Which airline has no coefficient of its own?

Exercise | Start Over | Show Solution | Run Code

```
1
```

One category is the base

$$\text{arr_delay}_i = \beta_0 + \beta_1\,\text{other}_i + \beta_2\,\text{us}_i + \beta_3\,\text{viva}_i + \beta_4\,\text{volaris}_i + u_i$$

carrier	Mean arrival delay	Coefficient
Aeromexico	-20.00	base
Other foreign	-9.62	10.38
US airline	-8.08	11.91
Viva Aerobus	-3.08	16.92
Volaris	-7.51	12.48

- Five categories need four 0/1 regressors. The one left out is the base; every coefficient is a gap to it.
- R leaves out the first category in alphabetical order: here, Aeromexico.

Why one category must be left out

Each flight belongs to exactly one airline, so the five dummies always add up to 1:

$$\text{aeromexico}_i + \text{other}_i + \text{us}_i + \text{viva}_i + \text{volaris}_i = 1$$

The column of 1s for the intercept is already in $\mathbf{X}$. What happens if all five dummies go in with it?

Exercise 3 — Five dummies

1. Build the five 0/1 columns, for example `flights$volaris <- (flights$carrier == "Volaris") * 1`.
2. Regress `flights$arr_delay` on four of them: `other`, `us`, `viva`, `volaris`. Compare with Exercise 2.
3. Add the fifth, `aeromexico`. What does `summary()` print for it? Why?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Are the gaps fair?

Aeromexico's flights land earliest. But the airlines may differ in more than the airline: how late their flights leave, and how long they fly.

Which airline lands latest for the same departure delay and the same scheduled time?

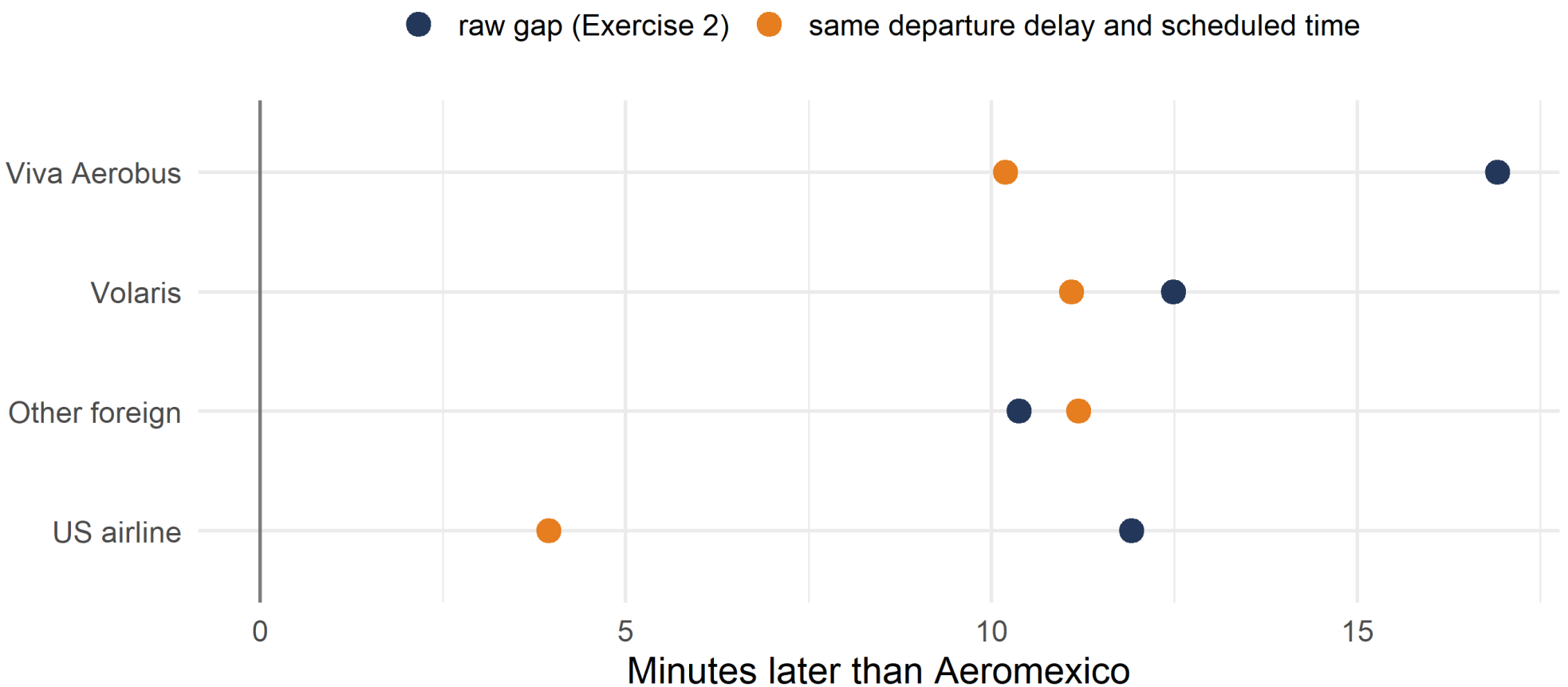
Exercise 4 — Same departure, same schedule

1. Regress `flights$arr_delay` on `flights$dep_delay`, `flights$sched_time` and `flights$carrier`.
2. Compare each airline's coefficient with Exercise 2. Which gaps shrink the most?
3. Use the mean departure delay of each airline to explain why.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Raw gap and fair gap



- For the same departure delay and scheduled time, Volaris lands 11.1 minutes later than Aeromexico and Viva Aerobus 10.2: Aeromexico builds more slack into its schedules, or makes up more time.
- The rule of week 10 holds for each dummy: a gap moves when a left-out cause differs across the groups.
- Volaris against Viva Aerobus? $11.10 - 10.19 = 0.90$ minutes. Whether that gap is significant needs week 12.

Another base, by hand

2 · Categories and fit

A category with many groups uses many degrees of freedom. Does it earn them?

Does the day of the week matter?

`flights$weekday` has seven categories; R takes Friday, first in alphabetical order, as the base. That is six dummies.

Which earns its place in the forecast of week 10: the day of the week or the airline?

Exercise 5 — Days, airlines and adjusted R^2

1. Regress `flights$arr_delay` on `flights$dep_delay` and `flights$sched_time`. Note the adjusted R^2.
2. Add `flights$weekday`. Note the adjusted R^2. Which days differ from Friday at 5%?
3. Replace `weekday` with `carrier`. Note the adjusted R^2.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Each group costs a degree of freedom

- A category with g groups adds $g - 1$ regressors. Adjusted R^2 charges for each one.
- The airline is worth its four: the forecast's typical miss falls from 13.4 to 12.5 minutes.
- The day of the week is not: six dummies for a typical miss of 13.4 minutes.

Do the six days matter together, even if most are not significant one by one? That is one test of several coefficients at once: week 12.

3 · Interactions

When the effect of one regressor depends on another.

New data: shelf prices from PROFECO

PROFECO inspectors record shelf prices in stores all over Mexico ("Quién es Quién en los Precios", open data). `drinks` has 498 stores in 30 states, November 2025 to February 2026. One row is one item in one store in one month (the median of the visits).

Column	What it is
`price`	shelf price, pesos
`litres`, `units`	litres in the pack; containers in it (1, 6 or 12)
`product`, `brand`, `diet`	soda, water or beer; the brand; 1 if a sugar-free soda
`format`, `chain`, `store`	store type; chain; the store
`local`	1 for a store outside the national chains: public markets, regional supermarkets, Neto, Circle K
`income`	municipal labour income per person, thousand pesos a month
`month`, `after`	the month; `after` is 1 from January 2026

Sources: PROFECO, Quién es Quién en los Precios; INEGI, Encuesta Intercensal 2025.

Is a litre worth the same in soda and in water?

`sw` holds the soft drinks and the bottled water of November 2025, one bottle at a time, up to 3 litres: 7,585 prices.

In week 10, a 0/1 regressor moved the whole line and kept its slope. Does an extra litre add the same number of pesos to a soda and to a bottle of water?

Exercise 6 — One slope or two?

1. Regress `price` on `litres` for the soda only, and then for the water only.
2. Create `sw$water`, 1 for water and 0 for soda. Regress `sw$price` on `sw$litres` and `sw$water`.
3. How many pesos per litre does each model give for soda? For water?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Two slopes in one regression

$$\text{price}_i = \beta_0 + \beta_1 \, \text{litres}_i + \beta_2 \, \text{water}_i + \beta_3 \, (\text{litres}_i \times \text{water}_i) + u_i$$

- For soda ($\text{water} = 0$): $\beta_0 + \beta_1 \, \text{litres}$.
- For water ($\text{water} = 1$): $(\beta_0 + \beta_2) + (\beta_1 + \beta_3) \, \text{litres}$.
- The product of the two regressors is one more column in $\mathbf{X}$. Build it with arithmetic: `sw$litres_water <- sw$litres * sw$water`.

Exercise 7 — Build the interaction

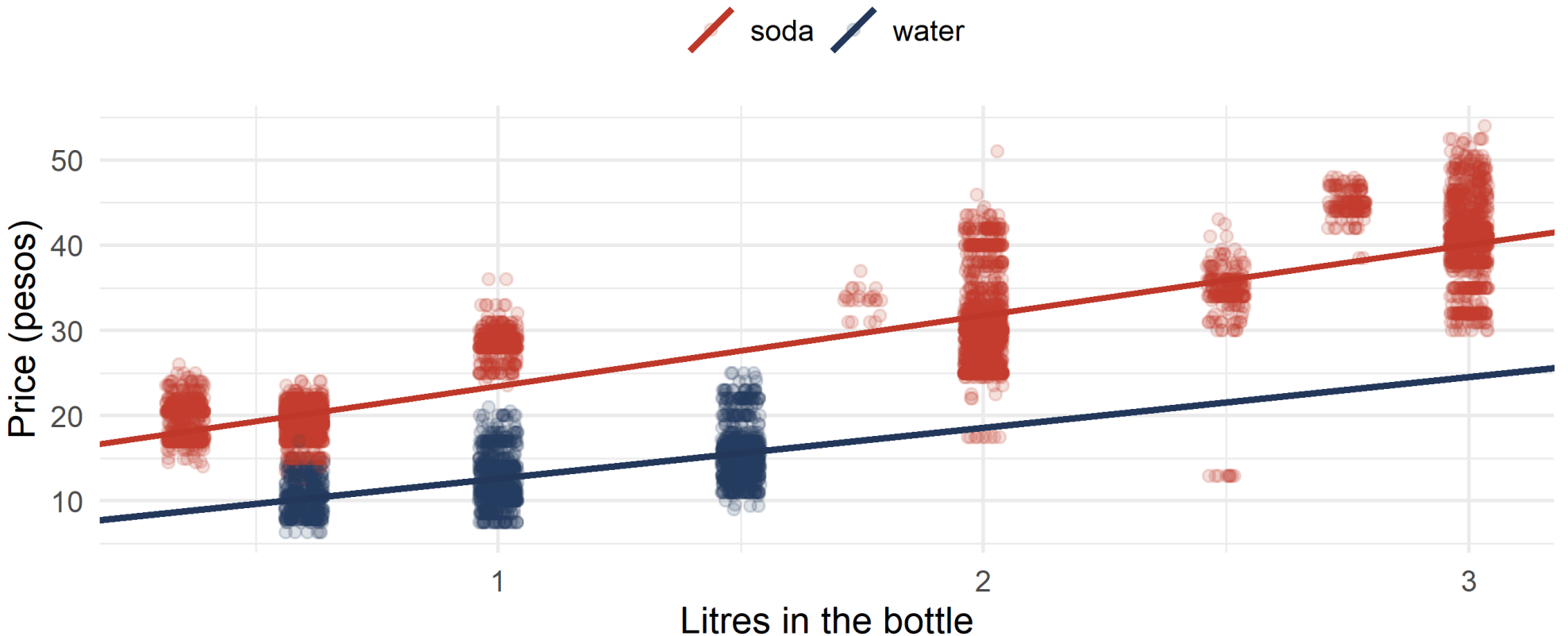
Does an extra litre add fewer pesos to water than to soda? By how many?

1. Create `sw$water` and `sw$litres_water`, the product of `sw$litres` and `sw$water`.
2. Regress `sw$price` on `sw$litres`, `sw$water` and `sw$litres_water`.
3. Recover the soda slope and the water slope of Exercise 6 from the coefficients.
4. Is the difference between the two slopes significant at 5%?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Lines that are not parallel



- A 0/1 regressor alone moves the line. Its product with `litres` turns the line.
- The interaction answers “does the effect of size depend on the product?” Here, yes: soda charges more for each extra litre.
- The same product works for two continuous regressors: the slope of one then changes with the level of the other.

Do local stores price to their neighbourhood?

Week 10 found that the national chains set national prices: municipal income barely moved the price of a soda.

`wat` holds the bottled water up to 1.5 litres, November 2025 to February 2026: 8,522 prices, 677 of them in local stores.

Does the price rise with municipal income in local stores, when it does not in the chains?

Exercise 8 — Local stores and income

1. Regress `wat$price` on `wat$litres` and `wat$income`.
2. Create `wat$income_local`, the product of `wat$income` and `wat$local`. Regress `wat$price` on `wat$litres`, `wat$income`, `wat$local` and `wat$income_local`.
3. How many pesos does 1,000 pesos more of municipal income add in a chain store? In a local store?
4. Above what income is a local store dearer than a chain store, for the same bottle?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Chains set one price; local stores follow the neighbourhood



- An interaction of a 0/1 regressor with a continuous one gives each group its own slope.
- For a bottler: in local stores the shelf price follows local income; in the chains it does not.

4 · Two groups, two periods

An interaction of two 0/1 regressors, and what a new tax did to prices.

The 2026 tax on sugary drinks

- On 1 January 2026 the tax (IEPS) on sugary drinks rose from 1.6451 to 3.0818 pesos per litre.
- Sugar-free soft drinks, untaxed until then, now pay 1.5 pesos per litre. Water pays nothing.
- VAT of 16% is charged on top of the tax. In a 600 ml bottle the extra tax is $0.6 \times (3.0818 - 1.6451) \times 1.16 = 1.00$ peso.

Did stores pass the tax on to the shelf price? Comparing December with January is not enough: prices change in January for other reasons too.

Water pays no tax but goes through the same January. Its change shows what would have happened to soda without the tax.

Exercise 9 — Did the price of a soda jump?

`tax600` holds the 600 ml bottles of soda and water in December 2025 and January 2026. `tax600$group` is “sugary soda”, “sugar-free soda” or “water”; `tax600$after` is 1 in January.

1. Compute the mean price of sugary soda and of water, in December and in January.
2. Keep the sugary soda and the water: `sw600 <- tax600[tax600$group != "sugar-free soda", ]`. Create `sw600$sugar` (1 for sugary soda) and `sw600$sugar_after`, the product of `sugar` and `after`.
3. Regress `sw600$price` on `sugar`, `after` and `sugar_after`. Match each coefficient to the four means.
4. Compare the coefficient on `sugar_after` with the extra tax in the bottle, 1.00 peso.

Exercise | Start Over | Show Solution | Run Code

```
1
```

The difference in differences

600 ml bottle	December	January	Change
sugary soda	19.32	21.04	1.71
water	10.63	10.93	0.29
difference	8.69	10.11	1.42

$$\text{price}_i = \beta_0 + \beta_1 \, \text{sugar}_i + \beta_2 \, \text{after}_i + \beta_3 \, (\text{sugar}_i \times \text{after}_i) + u_i$$

- Four groups, four coefficients: the regression reproduces the four means exactly.
- β_3 is a difference in differences: (soda's change) − (water's change).

Exercise 10 — Sugar-free soda

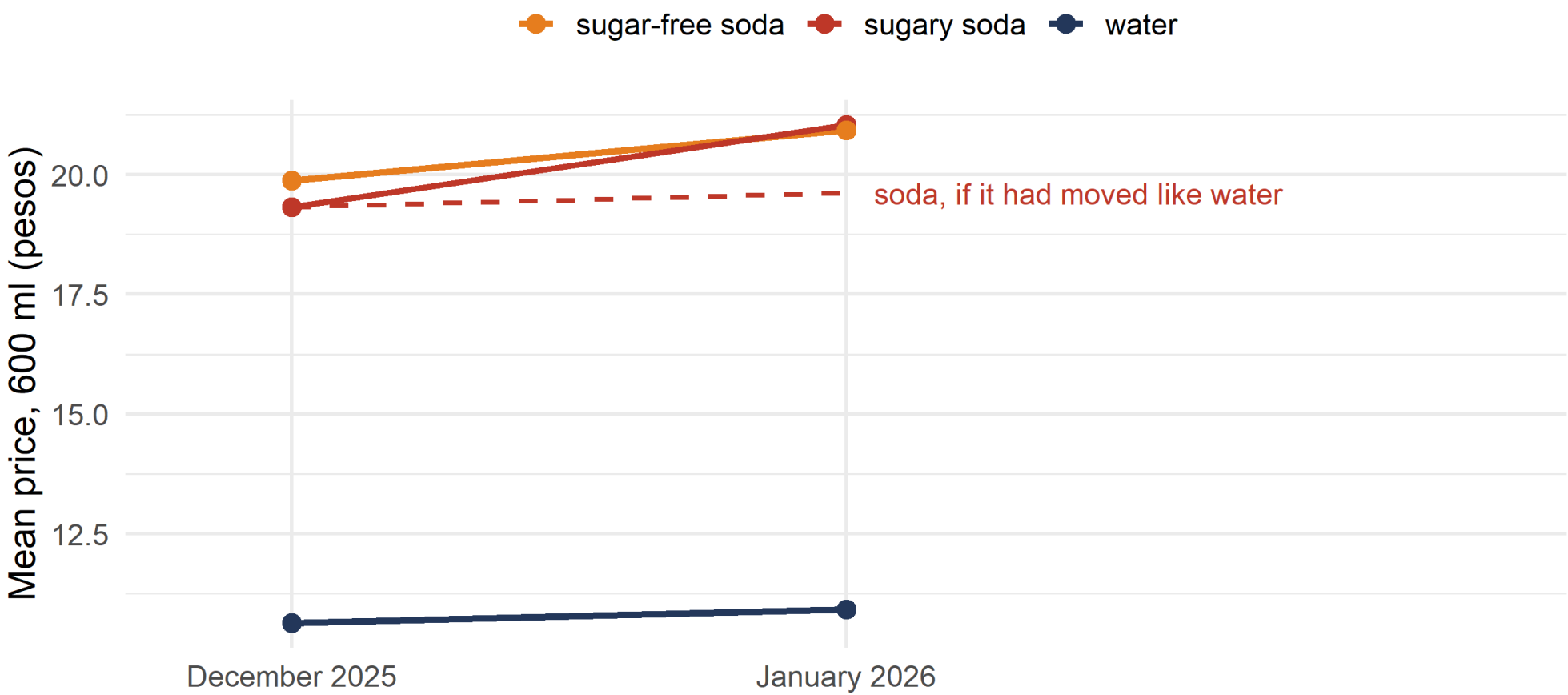
Sugar-free soda went from no tax to 1.5 pesos per litre: 1.04 peso in a 600 ml bottle, with VAT. Did the stores pass that on?

1. Keep the sugar-free soda and the water: `dw600 <- tax600[tax600$group != "sugary soda", ]`.
2. Build the 0/1 column for sugar-free soda and its product with `after`.
3. Regress the price on the three regressors and read the extra change for sugar-free soda.
4. Compare it with 1.04 peso, and with the sugary soda of Exercise 9.

Exercise | Start Over | Show Solution | Run Code

```
1
```

What the tax did



- The dashed line is the comparison: soda moving as water did. The gap at January is $\beta_3 = 1.42$ pesos.
- For a bottler or a retailer: the 2026 tax reached the shelf in full, and more, on sugary drinks.
- It only works if soda and water would have moved together without the tax. Water is the stand-in for what soda would have done.

What you should be able to do

- Put a regressor with several categories into `lm()` and read each coefficient as a gap to the base category.
- Say why one category must be left out.
- Explain why a gap between groups changes when other regressors are held fixed.
- Use adjusted R^2 to judge whether a category earns its dummies.
- Build an interaction with arithmetic and read each group's slope.
- Read an interaction of two 0/1 regressors as a difference in differences, from the four means or from the regression.

Next week: inference with several regressors. Is Volaris really later than Viva Aerobus? Do the six days matter together? How sure are we of a forecast?

Go to the exam practice →

Practice for this week

Weekly exercise set: `W11_exercises.pdf`, with solutions in `W11_solutions.pdf`.

You can already solve exercises 12 and 14, most of 8, 10 and 17, and 11 (b).

Not in these slides: 8 (f) and 9 use a 0/1 outcome, 9 (h) a squared regressor and 9 (i) standardized coefficients, which this course does not cover. 10 (d), 11 (a) and 16 compare two coefficients; 10 (h) and 11 (c) need intervals for a forecast; 11 (d)–(e), 13 and 15 test several coefficients at once or the assumptions: all week 12.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
8 (a)–(e)	Inflation forecasts, financial literacy and a college degree (`inflation_expectations`)	R with the data
10 (a)–(c), (e)–(g)	Weekly pay, children and marital status (`cps`)	R with the data
12	Reading an interaction of a 0/1 and a continuous regressor	By hand
17 (a)–(d)	Hours, pay and the gender gap (ENOE)	R with the data

Interactive versions: exam practice

Exam practice

Problems from this week's set, and a question from the Spring 2026 final. ← Back to the summary · ← Back to Practice for this week

Exam practice 1 — Who expects high inflation?

[P3 #8]

`inflation` holds 290 people in the United States, 2007–2008, from the `probstats4econ` package (`inflation_expectations`).

- `inflation$inflation_pred` — the inflation they expect next year, percentage points
- `inflation$age` — age, years
- `inflation$finlit_score` — financial literacy score, out of 12
- `inflation$collgrad` — 1 for a college graduate
- `inflation$famincome_hi` — 1 if family income is above 75,000 dollars

a. Regress `inflation_pred` on the four regressors. Which slopes are significant at 5%?

b. Interpret the slope on `finlit_score`.

c. Interpret the slope on `collgrad`.

d. Draw a histogram of the fitted values.

e. Add the product of `finlit_score` and `collgrad`. What is the p-value of the interaction? What is the slope on `finlit_score` for a college graduate, and for someone without a degree?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 2 — Pay, children and marriage

[P3 #10]

`cps` holds 4,013 people aged 30 to 59 from the 2019 Current Population Survey (`probstats4econ`). `cps$earnwk` is pay last week, dollars; `cps$gender` is “Female” or “Male”; `cps$ownchild` the number of own children at home; `cps$marstatus` has four categories; `cps$age` and `cps$educ` are in years.

a. Regress `earnwk` on `gender`. Interpret the two coefficients.

b. Add `ownchild`. What happens to the coefficient on `gender`?

c. Create `cps$female` (1 for women) and its product with `ownchild`. Regress `earnwk` on `female`, `ownchild` and the product. Interpret the coefficients.

d. Drop the product and add `marstatus`, `age` and `educ` to the model of (b). What happened to the `gender` coefficient?

e. How many dummies does `marstatus` use? Why?

f. Interpret the coefficient on the Married dummy.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 3 — Diet and exercise

[P3 #12]

100 people recorded their weight loss w (kg) after three months. $d = 1$ for those on a new diet; h is weekly hours of exercise. The fitted model is

$$\hat{w} = -1.4 + 7.2\,d + 1.5\,h + 3.4\,dh$$

Which statement is correct?

a. People who neither exercise nor follow the diet have a negative weight, which makes no sense.

b. People who follow the diet but do not exercise lose more than 5 kg.

c. For people not on the diet, one hour of exercise a week lowers weight by about 4.5 kg.

d. For people on the diet, one hour of exercise a week lowers weight by 10.7 kg.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 4 — Gym memberships

[P3 #14]

A gym has three memberships: limited, full and luxury. Muscle gain after three months is regressed on membership dummies and an intercept. The ANOVA table shows a total of 256 degrees of freedom and sums of squares of 500 (regression) and 1,200 (total).

What are the regression degrees of freedom and the sample size?

a. 2 and 257 (b) 3 and 257 (c) 4 and 256 (d) 2 and 254

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 5 — Hours, pay and the gender gap

[17]

`w12` holds the 2,000 ENOE workers of 2022 drawn in the set (Canvas, 02 Data, `ENOE_salaries.Rda`): `w12$Income` (monthly pesos), `w12$Hours_worked_lw` (hours last week) and `w12$Female` (1 for women).

a. Regress `Income` on `Hours_worked_lw` and `Female`. Report the three coefficients and R^2.

b. Interpret the coefficient on `Female`. What does this model assume about the slope on hours for the two groups?

c. Add the product of `Hours_worked_lw` and `Female`. Write the fitted line for men and for women.

d. Which group's pay rises faster with hours?

e. Is the difference in slopes significant at 5%? At 10%?

f. Write two sentences for an HR director.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Exam practice 6 — A tariff on cars

[Spring 2026 final, Q13–Q18]

A Mexican manufacturer exports cars and boats to the United States. In the second quarter of 2025 the US put a tariff on cars, not on boats. `exports` holds the quarterly sales of each product, 2002 to 2026 (simulated exam data): `exports$Sales`, `exports$car` (1 for cars) and `exports$after_q2_2025` (1 from 2025 Q2).

a. Match each combination of (`car`, `after_q2_2025`) — (0,0), (1,0), (0,1), (1,1) — to boats or cars, before or after the tariff.

b. Build the product of `car` and `after_q2_2025` and regress `Sales` on the three regressors. What are the predicted boat sales before the tariff? After?

c. Interpret the coefficient on `after_q2_2025`.

d. What are the predicted car sales before the tariff?

e. Interpret the coefficient on `car`.

f. What is the effect of the tariff on car sales?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Appendix — optional

Another base category, by hand.

Appendix — another base

To compare every airline with Volaris, leave Volaris out instead of Aeromexico. Build the dummies with arithmetic (as in Exercise 3) and choose which four go in:

```
reg <- lm(arr_delay ~ dep_delay + sched_time + aeromexico + other + us + viva,
          data = flights)
summary(reg)
```

Dummy	Minutes later than Volaris
aeromexico	-11.10
other	0.09
us	-7.15
viva	-0.90

- Every gap is now measured from Volaris. Viva Aerobus: -0.90, the same 10.19 − 11.10 as before.
- The fit, the slopes on `dep_delay` and `sched_time`, and every forecast are unchanged. Only the base moves.

← Back to the airlines